Burst Threshold Derivation in the Gauged Two-Gyro Hopf Lattice

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Abstract

The burst threshold governs when the local twist density $\Theta = 2\arccos(\operatorname{Re} q)$ triggers flux shedding in the gauged two-gyro Hopf lattice. We distinguish two related but numerically distinct quantities: (i) the *Hopf linking saturation* angle $\Theta_{\text{link}} = 2\pi W_g/(2W_g + 1) \approx \pi$ rad, and (ii) the *effective burst threshold* used in PDE and lattice simulations, $\theta_{\text{crit}} = \pi(1+\kappa) \approx 5.8$ rad with $\kappa \approx 0.85$. Earlier drafts incorrectly identified Θ_{link} with 5.8; this revision corrects that typo and relates both thresholds to S^3 geometry, the Hopf fibration, and global pointer holonomy. The reset rule and quantized winding $W_g = 350/\pi$ are unchanged.

1 Local Twist Density

The microscopic field is a unit quaternion $q(x,t) \in S^3$:

$$q = w + ix + jy + kz, \qquad w^2 + x^2 + y^2 + z^2 = 1.$$

The local twist density (Hopf-angle variable) is

$$\Theta(x,t) = 2\arccos(\operatorname{Re} q) = 2\arccos(w),$$

measuring accumulated angular momentum in the two-gyro counter-rotating helicities.

2 Two Thresholds: Geometry vs. Simulation

2.1 Antipodal fiber limit ($\Theta = \pi$)

On the Hopf fiber S^1, the antipodal point occurs when $w = \cos(\Theta/2) = 0$, hence

$$\Theta_{\text{antipodal}} = \pi \approx 3.1416 \text{ rad} \approx 180^\circ.$$

This is the bare half-turn limit on a single fiber.

2.2 Hopf linking saturation (Θ_{link})

For a map $q : T^3 \to S^3$, the Hopf invariant (linking number of preimages) is protected when local twist does not violate quantized linking. With locked winding $W_g = 350/\pi \approx 111.408$, stability requires

$$\boxed{\Theta_{\text{link}} = \frac{2\pi W_g}{2W_g + 1}} \tag{1}$$

Numerically, $\Theta_{\text{link}} \approx 3.128$ rad, coinciding with $\Theta_{\text{antipodal}}$ to $< 0.5\%$. **This is not** 5.8 **rad.** Equation (**??**) is the geometric linking bound; it saturates near π.

2.3 Effective burst threshold (θ_{crit})

The global pointer holonomy $\alpha = -\kappa\bar{\Theta}$ with $\kappa \approx 0.85$ and the Clifford-torus projection lift the *operational* burst threshold above the bare fiber limit. Continuum and lattice codes implement

$$\boxed{\theta_{\text{crit}} = \pi(1+\kappa) \approx 5.812 \text{ rad} \approx 333^\circ} \tag{2}$$

With $\kappa = 0.85$, $\theta_{\text{crit}} \approx 5.81$, matching the simulation default $\theta_{\text{crit}} \approx 5.8$ in `pde_relaxation.py` and lattice demos. Physically: the soliton tolerates twist up to π on the fiber, plus a holonomy-lifted margin $\kappa\pi$ before the 4th-order Skyrme term can no longer stabilize against drive $\Delta\omega$.

Symbol	Formula	Value (rad)	Role
$\Theta_{\text{antipodal}}$	π	3.142	Bare S^1 fiber half-turn
Θ_{link}	$2\pi W_g/(2W_g+1)$	3.128	Hopf linking saturation
θ_{crit}	$\pi(1+\kappa)$	5.812	PDE/lattice burst sink

Table 1: Reconciled burst thresholds with $W_g = 350/\pi$, $\kappa = 0.85$.

3 Burst Reset Rule

When $\Theta_i > \theta_{\text{crit}}$, the discrete rule resets

$$q_i \leftarrow \text{normalize}\big(0.3\,[1,0,0,0] + 0.7\,q_i\big), \qquad \Theta_i \leftarrow 0.15\,\Theta_i.$$

The factor 0.3 pulls the real part toward identity; 0.15 ensures post-burst twist lies below θ_{crit} while conserving Hopf charge modulo W_g. In continuum form,

$$U'(\Theta) = -B(\Theta) = -C(\Theta - \theta_{\text{crit}})_+^p, \qquad C \gg 1,\ p > 1.$$

4 Relation to Topological Invariants

Each burst sheds flux $\Delta\Phi \propto \Delta\omega/W_g$. The global winding $W_g = 350/\pi$ keeps total topological charge integer (or half-integer for fermions). Post-burst reset returns sites to the low-twist attractor while pointer α recenters the lattice, preserving braiding-phase cluster $\phi_b \in [0.765,\ 0.823]$.

5 Observer Synchronization

Embedded detectors rotate with the global pointer; observable burst strength is damped:

$$\langle \text{burst strength} \rangle \approx \frac{1}{\kappa\,\Delta t_{\text{burst}}} \ll 1.$$

6 Summary

- $\Theta_{\text{link}} = 2\pi W_g/(2W_g+1) \approx \pi$: Hopf linking / antipodal geometry.
- $\theta_{\text{crit}} = \pi(1+\kappa) \approx 5.8$: effective simulation threshold.
- Prior conflation of these two values was a notational error (≈ 5.8 attached to Eq. (**??**)); corrected in this revision.
- Reset rule, $W_g = 350/\pi$, and observer synchronization are unchanged.